

Implementation of change operations on workflow control-flow pattern based process models in BPMN language - Phase I & II

Pattern WCP-1 (Sequence)

Acceptance sequences ABC

Simple change operations

Insert

Since all activities have the same dependencies, we can consider the different cases for only one of the activities.

- T0 Insert activity X , condition: none - possible
- T1 Insert activity X , condition: $X \prec_d A$ - possible
- T2 Insert activity X , condition: $A \prec_d X$ - possible
- T3 Insert activity X , condition: $X \prec A$ - possible
- T4 Insert activity X , condition: $X \prec A$ - possible
- T5 Insert activity X , condition: $A \prec_d X$; $X \prec_d B$ - possible
- T6 Insert activity X , condition: $X \prec_d A$; $B \prec_d X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

- E0 Insert activity X , condition: none - possible
- E1 Insert activity X , condition: $A \Leftrightarrow X$ - possible
- E2 Insert activity X , condition: $A \nLeftrightarrow X$ - possible
- E3 Insert activity X , condition: $A \Rightarrow X$ - possible
- E4 Insert activity X , condition: $X \Rightarrow A$ - possible
- E5 Insert activity X , condition: $A \vee X$ - possible
- E6 Insert activity X , condition: $A \wedge X$ - possible
- E7 Insert activity X , condition: $A \Leftrightarrow X$; $B \Leftrightarrow X$ - possible
- E8 Insert activity X , condition: $A \Leftrightarrow X$; $B \nLeftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Remove

Remove activity A, B, C (all individual) - no problem

Modify

Modify $(A \Leftrightarrow B)$, new dependency type $(A \Rightarrow B)$ - possible (but does not lead to implication, but equivalence (dependency relaxation))

Modify $(A \Leftrightarrow B)$, new dependency type $(A \nLeftrightarrow B)$ - possible

Modify $(A \Leftrightarrow B)$, new dependency type $(A \wedge B)$ - possible

Modify $(A \Leftrightarrow B)$, new dependency type $(A \vee B)$ - possible (but does not lead to OR, but equivalence)

Modify $(A \Leftrightarrow B)$, new dependency type $(A - B)$ - possible (but does not lead to independence, but equivalence)

Modify $(A \prec_d B)$, new dependency type $(A \prec B)$ - possible (but does not lead to $\prec$, but $\prec_d$)

Modify $(A \prec_d B)$, new dependency type $(A \succ_d B)$ - possible (with console based application works; with BPC causes an error)

Modify $(A \prec C)$, new dependency type $(A \prec_d C)$ - not possible, no final matrix shown (seems to cause an error)

Combined change operations

Replace

Replace activity A with D - possible

Collapse

Collapse $\{A, B\}$ - possible

Collapse $\{A, B, C\}$ - possible

Collapse $\{A, C\}$ - Error: Activity B happens between the activities to be collapsed

De collapse

Decollapse activity A - works for every provided YAML file

Parallelize

Parallelize $\{A, B\}$ - possible

Parallelize $\{B, C\}$ - possible

Parallelize $\{A, C\}$ - Error: Activities happening in between

Parallelize $\{A, B, C\}$ - possible

Swap

Swap A with B - possible

Swap A with C - possible

Skip

Skip A (also for all other activities) - possible

Condition Update

Condition activity: A , depending activity C - possible (also for all other combinations)

Pattern WCP-2 (Parallel Split)

Acceptance sequences ABC ACB BAC BCA CAB CBA

Simple change operations

Insert

Since all activities have the same dependencies, we can consider the different cases for only one of the activities.

T0 Insert activity X , condition: none - possible

T1 Insert activity X , condition: $X \prec_d A$ - possible

T2 Insert activity X , condition: $A \prec_d X$ - possible

T3 Insert activity X , condition: $X \prec A$ - possible

T4 Insert activity X , condition: $X \prec A$ - possible

T5 Insert activity X , condition: $A \prec_d X$; $X \prec B$ - possible

T6 Insert activity X , condition: $X \prec_d A$; $X \prec_d B$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X , condition: none - possible

E1 Insert activity X , condition: $A \Leftrightarrow X$ - possible

E2 Insert activity X , condition: $A \nLeftrightarrow X$ - possible

E3 Insert activity X , condition: $A \Rightarrow X$ - possible

E4 Insert activity X , condition: $X \Rightarrow A$ - possible

E5 Insert activity X , condition: $A \vee X$ - possible

E6 Insert activity X , condition: $A \bar{\vee} X$ - possible

E7 Insert activity X , condition: $A \Leftrightarrow X$; $B \Leftrightarrow X$ - possible

E8 Insert activity X , condition: $A \Leftrightarrow X$; $B \nLeftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Remove

Remove A (also or all other activities) - possible

Modify

For all existential dependencies we have the same dependency type ($a_i \Leftrightarrow a_j$), so we only perform modify for one pair of activities

Modify ($A \Leftrightarrow B$), new dependency type ($A \Rightarrow B$) - possible (but does not lead to implication, but equivalence (dependency relaxation))

Modify ($A \Leftrightarrow B$), new dependency type ($A \nLeftrightarrow B$) - not possible, final result not displayed

Modify ($A \Leftrightarrow B$), new dependency type ($A \bar{\vee} B$) - not possible, final result not displayed

Modify ($A \Leftrightarrow B$), new dependency type ($A \vee B$) - possible (but does not lead to OR, but equivalence

Modify ($A \Leftrightarrow B$), new dependency type ($A - B$) - possible (but does not lead to independency, but equivalence)

For all temporal dependencies we have the same dependency type ($a_i - a_j$), so we only perform modify for one pair of activities. Modify ($A - B$), new dependency type ($A \prec_d B$) - possible

Modify ($A - B$), new dependency type ($A \prec B$) - possible

Combined change operations

Replace

Replace activity A (also for any other activity) - possible

Collapse

Collapse $\{A, B\}$ - possible

Collapse $\{A, B, C\}$ - possible

De-collapse

De-collapse A (also for any other activity and any matrix) - possible

Parallelize

Parallelize $\{A, B\}$ - possible (the BPC application shows an error, but with console only tool it works)

Parallelize $\{A, B, C\}$ - possible

Swap

Swap A and B (for any two activities) - possible

Skip

Skip activity A (for any activity) - possible

Condition Update

Condition activity A , depending activity B (also for any other combination) - possible

Pattern WCP-3 (Synchronization)

Acceptance sequences ABC BAC

Simple change operations

Insert

Activities A, B have the same dependency to activity C , so we only need to consider one of the activities from the pair. For the combined constraints, consider combinations inside the pair and with activity C .

T0 Insert activity X , condition: none - possible

T1 Insert activity X , condition: $X \prec_d A$ - possible

Insert activity X , condition: $X \prec_d C$ - possible

T2 Insert activity X , condition: $A \prec_d X$ - possible

Insert activity X , condition: $C \prec_d X$ - possible

T3 Insert activity X , condition: $X \prec A$ - possible

Insert activity X , condition: $X \prec C$ - possible

T4 Insert activity X , condition: $X \prec A$ - possible

Insert activity X , condition: $X \prec C$ - possible

T5 Insert activity X , condition: $A \prec_d X; B \prec X$ - possible

Insert activity X , condition: $A \prec_d X; C \prec X$ - possible

T6 Insert activity X , condition: $X \prec_d A; X \prec_d B$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

Insert activity X , condition: $X \prec_d A; X \prec_d C$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X , condition: none - possible

E1 Insert activity X , condition: $A \Leftrightarrow X$ - possible

Insert activity X , condition: $C \Leftrightarrow X$ - possible

E2 Insert activity X , condition: $A \nLeftrightarrow X$ - possible

Insert activity X , condition: $C \nLeftrightarrow X$ - possible

E3 Insert activity X , condition: $A \Rightarrow X$ - possible

Insert activity X , condition: $C \Rightarrow X$ - possible

E4 Insert activity X , condition: $X \Rightarrow A$ - possible

Insert activity X , condition: $X \Rightarrow C$ - possible

E5 Insert activity X , condition: $A \vee X$ - possible

Insert activity X , condition: $C \vee X$ - possible

E6 Insert activity X , condition: $A \wedge X$ - possible

Insert activity X , condition: $C \wedge X$ - possible

E7 Insert activity X , condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible

Insert activity X , condition: $A \Leftrightarrow X; C \Leftrightarrow X$ - possible

E8 Insert activity X , condition: $A \Leftrightarrow X; B \nLeftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Insert activity X , condition: $A \Leftrightarrow X; C \nLeftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Remove

Remove activity A (also for any other activity) - possible

Modify

The dependency between A, B is the same pattern as WCP-2, so we only consider the dependencies between A or B and activity C .

Modify ($A \Leftrightarrow C$), new dependency type ($A \Rightarrow C$) - possible (but does not lead to implication, but equivalence (dependency relaxation))

Modify ($A \Leftrightarrow C$), new dependency type ($A \Leftarrow C$) - possible (but does not lead to implication, but equivalence (dependency relaxation))

Modify $(A \Leftrightarrow C)$, new dependency type $(A \not\Leftarrow C)$ - not possible, does not display a final result
 Modify $(A \Leftrightarrow C)$, new dependency type $(A \bar{\wedge} C)$ - not possible, does not display a final result
 Modify $(A \Leftrightarrow C)$, new dependency type $(A \vee C)$ - possible (but does not lead to or, but equivalence)
 Modify $(A \Leftrightarrow C)$, new dependency type $(A - C)$ - possible (but does not lead to independence, but equivalence (dependency relaxation))

Modify $(A \prec C)$, new dependency type $(A \prec_d C)$ - possible
 Modify $(A \prec C)$, new dependency type $(A \succ C)$ - possible, but leads to direct temporal dependency
 Modify $(A \prec C)$, new dependency type $(A - C)$ - possible

Combined change operations

Replace

Replace activity A (also for any other activity) - possible

Collapse

Collapse $\{A, B\}$ - possible

Collapse $\{A, C\}$ - Error: Activity B happens between the activities to be collapsed

Collapse $\{A, B, C\}$ - possible

De collapse

De-collapse A (also for any other activity and any matrix) - possible

Parallelize

Parallelize $\{A, B\}$ - possible

Collapse $\{A, C\}$ - Error

Collapse $\{A, B, C\}$ - possible

Swap

Swap activity A with activity B (also for any other pair of activities) - possible

Skip

Skip activity A (also for any other activity) - possible

Condition Update

Condition update, condition activity A , depending activity C - possible

Pattern WCP-4 (Exclusive Choice)

Acceptance sequences A B

Simple change operations

Insert

Since all activities have the same dependencies, we can consider the different cases for only one of the activities.

- T0 Insert activity X , condition: none - possible
- T1 Insert activity X , condition: $X \prec_d A$ - possible
- T2 Insert activity X , condition: $A \prec_d X$ - possible
- T3 Insert activity X , condition: $X \prec A$ - possible
- T4 Insert activity X , condition: $X \prec A$ - possible
- T5 Insert activity X , condition: $A \prec_d X$; $X \prec B$ - possible
- T6 No contradiction possible

- E0 Insert activity X , condition: none - possible
- E1 Insert activity X , condition: $A \Leftrightarrow X$ - possible
- E2 Insert activity X , condition: $A \not\Leftrightarrow X$ - possible
- E3 Insert activity X , condition: $A \Rightarrow X$ - possible
- E4 Insert activity X , condition: $X \Rightarrow A$ - possible
- E5 Insert activity X , condition: $A \vee X$ - possible
- E6 Insert activity X , condition: $A \bar{\vee} X$ - possible
- E7 Insert activity X , condition: $A \Leftrightarrow X$; $B \not\Leftrightarrow X$ - possible
- E8 Insert activity X , condition: $A \Leftrightarrow X$; $B \Leftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Remove

- Remove A - possible
- Remove B - possible

Modify

- Modify ($A \not\Leftrightarrow B$), new dependency type ($A \Leftrightarrow B$) - possible
- Modify ($A \not\Leftrightarrow B$), new dependency type ($A \Leftarrow B$) - possible
- Modify ($A \not\Leftrightarrow B$), new dependency type ($A \bar{\vee} B$) - possible (but does not lead to NAND, but negated equivalence (dependency relaxation))
- Modify ($A \not\Leftrightarrow B$), new dependency type ($A \vee B$) - possible
- Modify ($A \not\Leftrightarrow B$), new dependency type ($A - B$) - possible (but does not lead to independence, but OR)

Modify ($A - B$), new dependency type ($A \prec B$) - possible (but does not lead to $\prec$, but independence)

Modify ($A - B$), new dependency type ($A \prec_d B$) - possible (but does not lead to $\prec$, but independence)

Combined change operations

Replace

- Replace A with C - possible

Collapse

- Collapse $\{A, B\}$ - possible

De collapse

- De-collapse A and B (any process to decollapse) - possible

Parallelize

- Parallelize $\{A, B\}$ - possible

Swap

- Swap activity A and B - possible

Skip

Skip activity A (also for activity B) - possible

Condition Update

Condition update: condition activity A , depending activity B - possible

Pattern WCP-5 (Simple Merge)

Acceptance sequences AD BD CD

Simple change operations

Insert

Activities A, B, C have the same dependency to activity D , so we only need to consider one of the activities from the pair. For the combined constraints, consider combinations inside the pair and with activity C .

T0 Insert activity X , condition: none - possible

T1 Insert activity X , condition: $X \prec_d A$ - possible

Insert activity X , condition: $X \prec_d D$ - possible

T2 Insert activity X , condition: $A \prec_d X$ - possible

Insert activity X , condition: $D \prec_d X$ - possible

T3 Insert activity X , condition: $X \prec A$ - possible

Insert activity X , condition: $X \prec D$ - possible

T4 Insert activity X , condition: $X \prec A$ - possible

Insert activity X , condition: $X \prec D$ - possible

T5 Insert activity X , condition: $A \prec_d X; B \prec_d X$ - possible

Insert activity X , condition: $A \prec_d X; X \prec D$ - possible

T6 Insert activity X , condition: $X \prec_d A; D \prec_d X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X , condition: none - possible

E1 Insert activity X , condition: $A \Leftrightarrow X$ - possible

Insert activity X , condition: $D \Leftrightarrow X$ - possible

E2 Insert activity X , condition: $A \nLeftrightarrow X$ - possible

Insert activity X , condition: $D \nLeftrightarrow X$ - possible

E3 Insert activity X , condition: $A \Rightarrow X$ - possible

Insert activity X , condition: $D \Rightarrow X$ - possible

E4 Insert activity X , condition: $X \Rightarrow A$ - possible

Insert activity X , condition: $X \Rightarrow D$ - possible

E5 Insert activity X , condition: $A \vee X$ - possible

Insert activity X , condition: $D \vee X$ - possible

E6 Insert activity X , condition: $A \bar{\wedge} X$ - possible

Insert activity X , condition: $D \bar{\wedge} X$ - possible

E7 Insert activity X , condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible

Insert activity X , condition: $A \Leftrightarrow X; D \Leftrightarrow X$ - possible

E8 No contradiction

Remove

Remove activity A (for any activity) - possible

Modify

Activities A, B, C have the same dependencies to each other, so we only consider one of the pairs. The dependencies to activity D are the same for A, B, C , so we only consider one of the pairs here.

Modify $(A \bar{\wedge} B)$, new dependency type $(A \Leftrightarrow B)$ - possible (but A, B no longer part of the result (contradiction))

Modify $(A \bar{\wedge} B)$, new dependency type $(A \Rightarrow B)$ - possible (but A no longer part of the result (contradiction))

Modify $(A \bar{\wedge} B)$, new dependency type $(A \nLeftrightarrow B)$ - possible (but C no longer part of the result (contradiction))

Modify $(A \bar{\wedge} B)$, new dependency type $(A \vee B)$ - possible (but C no longer part of the result (contradiction))

Modify $(A \bar{\wedge} B)$, new dependency type $(A - B)$ - possible (but NAND instead of independence)

Modify $(A \Rightarrow D)$, new dependency type $(A \Leftrightarrow D)$ - possible (but B, C no longer part of the result (contradiction))

Modify $(A \Rightarrow D)$, new dependency type $(A \Leftarrow D)$ - possible (but B, C no longer part of the result (contradiction))

Modify $(A \Rightarrow D)$, new dependency type $(A \nRightarrow D)$ - possible

Modify $(A \Rightarrow D)$, new dependency type $(A \overline{\wedge} D)$ - possible (negated equivalence instead of NAND)

Modify $(A \Rightarrow D)$, new dependency type $(A \vee D)$ - possible

Modify $(A \Rightarrow D)$, new dependency type $(A - D)$ - possible (OR instead of independence)

Modify $(A - B)$, new dependency type $(A \prec B)$ - possible (but independence instead of $\prec$)

Modify $(A - B)$, new dependency type $(A \prec_d B)$ - possible (but independence instead of $\prec$)

Modify $(A \prec_d D)$, new dependency type $(A \prec D)$ - possible (but $\prec_d$ instead of $\prec$)

Modify $(A \prec_d D)$, new dependency type $(A \succ D)$ - possible (but $\succ_d$ instead of $\succ$)

Modify $(A \prec_d D)$, new dependency type $(A - D)$ - possible

Combined change operations

Replace

Replace activity A (also for any other activity) - possible

Collapse

Collapse $\{A, B\}$ - possible

Collapse $\{A, D\}$ - possible

Collapse $\{A, B, C\}$ - possible

De collapse

De-collapse A (for any other activity and any process to de-collapse) - possible

Parallelize

Parallelize $\{A, B\}$ - possible

Parallelize $\{A, D\}$ - possible

Parallelize $\{A, B, C, D\}$ - possible

Swap

Swap activity A and B (also for any pair of activities) - possible

Skip

Skip activity A - possible

Skip activity D - possible

Condition Update

Condition update, condition activity A , depending activity D - possible

Condition update, condition activity A , depending activity B - possible

Pattern WCP-6 (Multi-Choice)

Acceptance sequences AB AC AD ABC ABD ACB ACD ADB ADC ABCD ABDC ACBD ACDB
ADBC ADCB

Simple change operations

Insert

Activities B, C, D have the same dependency to activity A , so we only need to consider one of the activities from the set. For the combined constraints, consider combinations inside the pair and with activity A .

T0 Insert activity X , condition: none - possible

T1 Insert activity X , condition: $X \prec_d A$ - possible

Insert activity X , condition: $X \prec_d B$ - possible

T2 Insert activity X , condition: $A \prec_d X$ - possible

Insert activity X , condition: $B \prec_d X$ - possible

T3 Insert activity X , condition: $X \prec A$ - possible

Insert activity X , condition: $X \prec B$ - possible

T4 Insert activity X , condition: $X \prec A$ - possible

Insert activity X , condition: $X \prec B$ - possible

T5 Insert activity X , condition: $A \prec_d X; X \prec B$ - possible

Insert activity X , condition: $B \prec_d X; C \prec X$ - possible

T6 Insert activity X , condition: $X \prec_d A; B \prec_d X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

Insert activity X , condition: $X \prec_d C; X \prec_d B$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X , condition: none - possible

E1 Insert activity X , condition: $A \Leftrightarrow X$ - possible

Insert activity X , condition: $B \Leftrightarrow X$ - possible

E2 Insert activity X , condition: $A \nLeftrightarrow X$ - possible

Insert activity X , condition: $B \nLeftrightarrow X$ - possible

E3 Insert activity X , condition: $A \Rightarrow X$ - possible

Insert activity X , condition: $B \Rightarrow X$ - possible

E4 Insert activity X , condition: $X \Rightarrow A$ - possible

Insert activity X , condition: $X \Rightarrow B$ - possible

E5 Insert activity X , condition: $A \vee X$ - possible

Insert activity X , condition: $B \vee X$ - possible

E6 Insert activity X , condition: $A \wedge X$ - possible

Insert activity X , condition: $B \wedge X$ - possible

E7 Insert activity X , condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible

Insert activity X , condition: $B \Leftrightarrow X; C \nLeftrightarrow X$ - possible

E8 Insert activity X , condition: $A \Leftrightarrow X; B \nLeftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Remove

Remove activity A (also for any other activity) - possible

Modify

Activities B, C, D all have the same dependencies to each other, so must only consider one pair of them. Additionally they share the same dependency to activity A , so we only consider the pair (A, B) .

Modify $(A \Rightarrow B)$, new dependency type $(A \Leftrightarrow B)$ - possible

Modify $(A \Leftarrow B)$, new dependency type $(A \Rightarrow B)$ - possible

Modify $(A \Rightarrow B)$, new dependency type $(A \not\Rightarrow B)$ - possible

$(A \Leftarrow B)$ - possible

Modify $(A \Rightarrow B)$, new dependency type $(A \wedge B)$ - possible (instead of NAND we get negated equiva-

lence)

Modify $(A \Rightarrow B)$, new dependency type $(A \vee B)$ - possible

Modify $(A \Rightarrow B)$, new dependency type $(A - B)$ - possible (instead of independence we get OR)

Modify $(B - C)$, new dependency type $(B \Leftrightarrow C)$ - possible

Modify $(B - C)$, new dependency type $(B \Rightarrow C)$ - possible (instead of implication we get equivalence (dependency relaxation))

Modify $(B - C)$, new dependency type $(B \not\Rightarrow C)$ - possible

Modify $(B - C)$, new dependency type $(B \wedge C)$ - possible

Modify $(B - C)$, new dependency type $(B \vee C)$ - possible

Modify $(A \prec B)$, new dependency type $(A \prec_d B)$ - possible

Modify $(A \prec B)$, new dependency type $(A \succ B)$ - possible (direct temporal dependency instead of eventual)

Modify $(A \prec B)$, new dependency type $(A - B)$ - possible

Combined change operations

Replace

Replace activity A (also for any other activity) - possible

Collapse

Collapse $\{A, B\}$ - not possible, activities happening in between

Collapse $\{B, C\}$ - possible

Collapse $\{B, C, D\}$ - possible

Collapse $\{A, B, C, D\}$ - possible

De collapse

De-collapse activity A (also any other activity for any collapsed process) - possible

Parallelize

Parallelize $\{A, B\}$ - not possible, activities happening in between

Parallelize $\{B, C\}$ - possible

Parallelize $\{B, C, D\}$ - possible

Parallelize $\{A, B, C, D\}$ - possible

Swap

Swap activity A with B (or any other combination) - possible

Skip

Skip activity A (or any other activity) - possible

Condition Update

Condition update, condition activity A , depending activity B (or any other pair of activities) - possible

Pattern WCP-7 (Structured Synchronizing Merge)

Acceptance sequences AD BD CD ABD ACD BCD BAD CAD CBD ABCD ACBD BACD BCAD CABD CBAD

Simple change operations

Insert

Activities A, B, C have the same dependency to activity D , so we only need to consider one of the activities from the pair. For the combined constraints, consider combinations inside the pair and with activity C .

- T0 Insert activity X , condition: none - possible
- T1 Insert activity X , condition: $X \prec_d A$ - possible
- Insert activity X , condition: $X \prec_d D$ - possible
- T2 Insert activity X , condition: $A \prec_d X$ - possible
- Insert activity X , condition: $D \prec_d X$ - possible
- T3 Insert activity X , condition: $X \prec A$ - possible
- Insert activity X , condition: $X \prec D$ - possible
- T4 Insert activity X , condition: $X \prec A$ - possible
- Insert activity X , condition: $X \prec D$ - possible
- T5 Insert activity X , condition: $A \prec_d X; X \prec B$ - possible
- Insert activity X , condition: $A \prec_d X; X \prec D$ - possible
- T6 Insert activity X , condition: $X \prec_d A; D \prec_d X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

- E0 Insert activity X , condition: none - possible
- E1 Insert activity X , condition: $A \Leftrightarrow X$ - possible
- Insert activity X , condition: $D \Leftrightarrow X$ - possible
- E2 Insert activity X , condition: $A \nLeftrightarrow X$ - possible
- Insert activity X , condition: $D \nLeftrightarrow X$ - possible
- E3 Insert activity X , condition: $A \Rightarrow X$ - possible
- Insert activity X , condition: $D \Rightarrow X$ - possible
- E4 Insert activity X , condition: $X \Rightarrow A$ - possible
- Insert activity X , condition: $X \Rightarrow D$ - possible
- E5 Insert activity X , condition: $A \vee X$ - possible
- Insert activity X , condition: $D \vee X$ - possible
- E6 Insert activity X , condition: $A \wedge X$ - possible
- Insert activity X , condition: $D \wedge X$ - possible
- E7 Insert activity X , condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible
- Insert activity X , condition: $A \Leftrightarrow X; D \Leftrightarrow X$ - possible
- E8 No contradiction

Remove

Remove activity A (also for any other activity) - possible

Modify

Activities A, B, C all have the same dependencies to each other, so must only consider one pair of them. Additionally they share the same dependency to activity D , so we only consider the pair (A, B) .

- Modify $(C \Rightarrow D)$, new dependency type $(C \Leftrightarrow D)$ - possible
- Modify $(C \Rightarrow D)$, new dependency type $(C \Leftarrow D)$ - possible
- Modify $(C \Rightarrow D)$, new dependency type $(C \nLeftarrow D)$ - possible
- Modify $(C \Rightarrow D)$, new dependency type $(C \nrightarrow D)$ - possible (instead of NAND we get negated equivalence)
- Modify $(C \Rightarrow D)$, new dependency type $(C \vee D)$ - possible
- Modify $(C \Rightarrow D)$, new dependency type $(C - D)$ - possible (instead of independence we get OR)

Modify $(B - C)$, new dependency type $(B \Leftrightarrow C)$ - possible
 Modify $(B - C)$, new dependency type $(B \Rightarrow C)$ - possible
 Modify $(B - C)$, new dependency type $(B \nRightarrow C)$ - possible
 Modify $(B - C)$, new dependency type $(B \wedge C)$ - possible
 Modify $(B - C)$, new dependency type $(B \vee C)$ - possible

Modify $(C \prec D)$, new dependency type $(C \prec_d D)$ - possible
 Modify $(C \prec D)$, new dependency type $(C \succ D)$ - possible (direct temporal dependency instead of eventual)
 Modify $(C \prec D)$, new dependency type $(C - D)$ - possible

Modify $(B - C)$, new dependency type $(B \prec C)$ - possible
 Modify $(B - C)$, new dependency type $(B \prec_d C)$ - possible

Combined change operations

Replace

Replace activity A (also for any other activity) - possible

Collapse

Collapse $\{A, B\}$ - possible
 Collapse $\{A, B, C\}$ - possible
 Collapse $\{A, B, D\}$ - Error: Activity C happens between the activities to be collapsed
 Collapse $\{A, B, C, D\}$ - possible

De collapse

De-collapse activity A (also for any other activity with any collapsed process) - possible

Parallelize

Parallelize $\{A, B, C\}$ - possible
 Parallelize $\{A, B, D\}$ - Error
 Parallelize $\{A, B, C, D\}$ - possible

Swap

Swap activity A with activity D (also for any pair of activities) - possible

Skip

Skip activity A (also for any other activity) - possible

Condition Update

Condition update, condition activity A , depending activity B (also for any other pair of activities)-possible

Pattern WCP-10 (Arbitrary Cycles)

Acceptance sequences ABC ABCBC ABCBCBC ... C BC BCBC BCBCBC ...

Simple change operations

Insert

T0 Insert activity X , condition: none - possible
T1 Insert activity X , condition: $X \prec_d A$ - possible
Insert activity X , condition: $X \prec_d B$ - possible
Insert activity X , condition: $X \prec_d C$ - possible
T2 Insert activity X , condition: $A \prec_d X$ - possible
Insert activity X , condition: $B \prec_d X$ - possible
Insert activity X , condition: $C \prec_d X$ - possible
T3 Insert activity X , condition: $X \prec A$ - possible
Insert activity X , condition: $X \prec B$ - possible
Insert activity X , condition: $X \prec C$ - possible
T4 Insert activity X , condition: $X \prec A$ - possible
Insert activity X , condition: $X \prec B$ - possible
Insert activity X , condition: $X \prec C$ - possible
T5 Insert activity X , condition: $A \prec_d X; B \prec X$ - possible
Insert activity X , condition: $A \prec_d X; X \prec B$ - possible
T6 Insert activity X , condition: $X \prec_d A; B \prec X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X , condition: none - possible
E1 Insert activity X , condition: $A \Leftrightarrow X$ - possible
Insert activity X , condition: $B \Leftrightarrow X$ - possible
Insert activity X , condition: $C \Leftrightarrow X$ - possible
E2 Insert activity X , condition: $A \nLeftrightarrow X$ - possible
Insert activity X , condition: $B \nLeftrightarrow X$ - possible
Insert activity X , condition: $C \nLeftrightarrow X$ - possible
E3 Insert activity X , condition: $A \Rightarrow X$ - possible
Insert activity X , condition: $B \Rightarrow X$ - possible
Insert activity X , condition: $C \Rightarrow X$ - possible
E4 Insert activity X , condition: $X \Rightarrow A$ - possible
Insert activity X , condition: $X \Rightarrow B$ - possible
Insert activity X , condition: $X \Rightarrow C$ - possible
E5 Insert activity X , condition: $A \vee X$ - possible
Insert activity X , condition: $B \vee X$ - possible
Insert activity X , condition: $C \vee X$ - possible
E6 Insert activity X , condition: $A \wedge X$ - possible
Insert activity X , condition: $B \wedge X$ - possible
Insert activity X , condition: $C \wedge X$ - possible
E7 Insert activity X , condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible
E8 No contradiction

Remove

Remove activity A - possible Remove activity B - possible Remove activity C - possible

Modify

Modify ($A \Rightarrow B$), new dependency type ($A \Leftrightarrow B$) - possible
Modify ($A \Rightarrow B$), new dependency type ($A \Leftarrow B$) - possible
Modify ($A \Rightarrow B$), new dependency type ($A \nLeftrightarrow B$) - possible
Modify ($A \Rightarrow B$), new dependency type ($A \wedge B$) - possible
Modify ($A \Rightarrow B$), new dependency type ($A \vee B$) - possible
Modify ($A \Rightarrow B$), new dependency type ($A - B$) - possible

Modify $(A \Rightarrow C)$, new dependency type $(A \Leftrightarrow C)$ - possible
 Modify $(A \Rightarrow C)$, new dependency type $(A \Leftarrow C)$ - possible
 Modify $(A \Rightarrow C)$, new dependency type $(A \nRightarrow C)$ - possible (A no longer in the matrix (contradiction))
 Modify $(A \Rightarrow C)$, new dependency type $(A \vee C)$ - possible

Modify $(B \Rightarrow C)$, new dependency type $(B \Leftrightarrow C)$ - possible
 Modify $(B \Rightarrow C)$, new dependency type $(B \Leftarrow C)$ - possible
 Modify $(B \Rightarrow C)$, new dependency type $(B \nRightarrow C)$ - possible (A no longer part of the process (contradiction))
 Modify $(B \Rightarrow C)$, new dependency type $(B \wedge C)$ - possible (negated existential equivalence instead of NAND, A no longer part of the process)
 Modify $(B \Rightarrow C)$, new dependency type $(B - C)$ - possible (OR instead of independence)

Modify $(A \prec_d B)$, new dependency type $(A \prec B)$ - possible (direct temporal dependency instead of eventual)
 Modify $(A \prec_d B)$, new dependency type $(A \succ B)$ - possible (A no longer part of the process (contradiction))
 Modify $(A \prec_d B)$, new dependency type $(A - B)$ - possible (direct temporal dependency instead of independence)

Modify $(A \prec C)$, new dependency type $(A \prec_d C)$ - possible (A no longer part of the process (contradiction))
 Modify $(A \prec C)$, new dependency type $(A \succ C)$ - possible (A no longer part of the process (contradiction))
 Modify $(A \prec C)$, new dependency type $(A - C)$ - possible (eventual temporal dependency instead of independence)

Modify $(B \prec_d C)$, new dependency type $(B \prec C)$ - possible (direct instead of eventual)
 Modify $(B \prec_d C)$, new dependency type $(B \succ C)$ - possible (A no longer part of the process (contradiction))
 Modify $(B \prec_d C)$, new dependency type $(B - C)$ - possible

Combined change operations

Replace

Replace activity A (also possible for any other activity) with X - possible

Collapse

Collapse $\{A, B\}$ - possible
 Collapse $\{A, C\}$ - Error: Activity B happens between the activities to be collapsed
 Collapse $\{B, C\}$ - possible
 Collapse $\{A, B, C\}$ - possible

De collapse

De-collapse activity A (also for any other activity with any collapsed process) - possible

Parallelize

Parallelize $\{A, B\}$ - possible
 Parallelize $\{A, C\}$ - Error: Activity B happens between the activities to be parallelized
 Collapse $\{B, C\}$ - possible
 Collapse $\{A, B, C\}$ - possible

Swap

Swap activity A with C (also for any other pair of activities) - possible

Skip

Skip activity A (also for any other activity) - possible

Condition Update

Condition update, condition activity A , depending activity B (also for any other pair of activities)-possible

Pattern WCP-17 (Interleaved Parallel Routing)

Acceptance sequences ABC ACB CAB

Simple change operations

Insert

T0 Insert activity X , condition: none - possible
T1 Insert activity X , condition: $X \prec_d A$ - possible
Insert activity X , condition: $X \prec_d B$ - possible
Insert activity X , condition: $X \prec_d C$ - possible
T2 Insert activity X , condition: $A \prec_d X$ - possible
Insert activity X , condition: $B \prec_d X$ - possible
Insert activity X , condition: $C \prec_d X$ - possible
T3 Insert activity X , condition: $X \prec A$ - possible
Insert activity X , condition: $X \prec B$ - possible
Insert activity X , condition: $X \prec C$ - possible
T4 Insert activity X , condition: $X \prec A$ - possible
Insert activity X , condition: $X \prec B$ - possible
Insert activity X , condition: $X \prec C$ - possible
T5 Insert activity X , condition: $A \prec_d X; B \prec X$ - possible
Insert activity X , condition: $A \prec_d X; X \prec B$ - possible
T6 Insert activity X , condition: $X \prec_d A; B \prec X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X , condition: none - possible
E1 Insert activity X , condition: $A \Leftrightarrow X$ - possible
Insert activity X , condition: $B \Leftrightarrow X$ - possible
Insert activity X , condition: $C \Leftrightarrow X$ - possible
E2 Insert activity X , condition: $A \nLeftrightarrow X$ - possible
Insert activity X , condition: $B \nLeftrightarrow X$ - possible
Insert activity X , condition: $C \nLeftrightarrow X$ - possible
E3 Insert activity X , condition: $A \Rightarrow X$ - possible
Insert activity X , condition: $B \Rightarrow X$ - possible
Insert activity X , condition: $C \Rightarrow X$ - possible
E4 Insert activity X , condition: $X \Rightarrow A$ - possible
Insert activity X , condition: $X \Rightarrow B$ - possible
Insert activity X , condition: $X \Rightarrow C$ - possible
E5 Insert activity X , condition: $A \vee X$ - possible
Insert activity X , condition: $B \vee X$ - possible
Insert activity X , condition: $C \vee X$ - possible
E6 Insert activity X , condition: $A \wedge X$ - possible
Insert activity X , condition: $B \wedge X$ - possible
Insert activity X , condition: $C \wedge X$ - possible
E7 Insert activity X , condition: $A \Leftrightarrow X; B \Leftrightarrow X$ - possible
E8 Insert activity X , condition: $A \Leftrightarrow X, C \nLeftrightarrow X$ - Error: The input is invalid: Existential dependencies cause a contradiction.

Remove

Remove activity A - possible
Remove activity B - possible
Remove activity C - possible

Modify

The pairs of activities (A, C) and (B, C) have the same dependencies, so we must only consider one of the pairs.

Modify $(A \Leftrightarrow B)$, new dependency type $(A \Rightarrow B)$ - possible (existential equivalence instead of

implication (dependency relaxation))

Modify $(A \Leftrightarrow B)$, new dependency type $(A \Leftarrow B)$ - possible (existential equivalence instead of implication (dependency relaxation))

Modify $(A \Leftrightarrow B)$, new dependency type $(A \nLeftarrow B)$ - not possible, contradiction

Modify $(A \Leftrightarrow B)$, new dependency type $(A \overline{\Leftarrow} B)$ - not possible, contradiction

Modify $(A \Leftrightarrow B)$, new dependency type $(A \vee B)$ - possible (existential equivalence instead of OR)

Modify $(A \Leftrightarrow B)$, new dependency type $(A - B)$ - possible (existential equivalence instead of independence)

Modify $(A \Leftrightarrow C)$, new dependency type $(A \Rightarrow C)$ - possible (existential equivalence instead of implication (dependency relaxation))

Modify $(A \Leftrightarrow C)$, new dependency type $(A \Leftarrow C)$ - possible (existential equivalence instead of implication (dependency relaxation))

Modify $(A \Leftrightarrow C)$, new dependency type $(A \nLeftarrow C)$ - not possible, contradiction

Modify $(A \Leftrightarrow C)$, new dependency type $(A \overline{\Leftarrow} C)$ - not possible, contradiction

Modify $(A \Leftrightarrow C)$, new dependency type $(A \vee C)$ - possible (existential equivalence instead of OR)

Modify $(A \Leftrightarrow C)$, new dependency type $(A - C)$ - possible (existential equivalence instead of independence)

Modify $(A \prec B)$, new dependency type $(A \prec_d B)$ - possible

Modify $(A \prec B)$, new dependency type $(A \succ B)$ - possible

Modify $(A \prec B)$, new dependency type $(A - B)$ - possible

Modify $(A - C)$, new dependency type $(A \prec C)$ - possible

Modify $(A - C)$, new dependency type $(A \prec_d C)$ - possible

Modify $(A - C)$, new dependency type $(A \succ C)$ - possible (direct temporal dependency instead of eventual)

Modify $(A - C)$, new dependency type $(A \succ_d C)$ - possible

Combined change operations

Replace

Replace activity A (also for any other activity) - possible

Collapse

Collapse $\{A, B\}$ - possible

Collapse $\{A, C\}$ - Error: Activity B happens between the activities to be collapsed

Collapse $\{B, C\}$ - Error: Activity A happens between the activities to be collapsed

Collapse $\{A, B, C\}$ - possible

De collapse

De-collapse activity A (also for any other activity; with any collapsed process) - possible

Parallelize

Parallelize $\{A, B\}$ - possible

Parallelize $\{A, C\}$ - Error

Parallelize $\{B, C\}$ - Error

Parallelize $\{A, B, C\}$ - possible

Swap

Swap activity A with B (also for any other combination) - possible

Skip

Skip activity A - possible

Skip activity B - possible

Skip activity C - possible

Condition Update

Condition update, condition activity A , depending activity B - possible

Condition update, condition activity A , depending activity C - possible

Condition update, condition activity B , depending activity C - possible

Condition update, condition activity B , depending activity A - possible

Condition update, condition activity C , depending activity A - possible

Condition update, condition activity C , depending activity B - possible

Pattern WCP-19 (Cancel Activity)

Acceptance sequences AB A

Simple change operations

Insert

T0 Insert activity X , condition: none - possible
T1 Insert activity X , condition: $X \prec_d A$ - possible
Insert activity X , condition: $X \prec_d B$ - possible
T2 Insert activity X , condition: $A \prec_d X$ - possible
Insert activity X , condition: $B \prec_d X$ - possible
T3 Insert activity X , condition: $X \prec A$ - possible
Insert activity X , condition: $X \prec B$ - possible
T4 Insert activity X , condition: $X \prec A$ - possible
Insert activity X , condition: $X \prec B$ - possible
T5 Insert activity X , condition: $A \prec_d X$; $X \prec B$ - possible
T6 Insert activity X , condition: $A \prec_d X$, $B \prec_d X$ - Error: The input is invalid: Temporal dependencies cause a contradiction.

E0 Insert activity X , condition: none - possible
E1 Insert activity X , condition: $A \Leftrightarrow X$ - possible
Insert activity X , condition: $B \Leftrightarrow X$ - possible
E2 Insert activity X , condition: $A \nLeftrightarrow X$ - possible
Insert activity X , condition: $B \nLeftrightarrow X$ - possible
E3 Insert activity X , condition: $A \Rightarrow X$ - possible
Insert activity X , condition: $B \Rightarrow X$ - possible
E4 Insert activity X , condition: $X \Rightarrow A$ - possible
Insert activity X , condition: $X \Rightarrow B$ - possible
E5 Insert activity X , condition: $A \vee X$ - possible
Insert activity X , condition: $B \vee X$ - possible
E6 Insert activity X , condition: $A \bar{\wedge} X$ - possible
Insert activity X , condition: $B \bar{\wedge} X$ - possible
E7 Insert activity X , condition: $A \Leftrightarrow X$; $B \Leftrightarrow X$ - possible
E8 No contradiction

Remove

Remove activity A - possible
Remove activity B - possible

Modify

Modify ($A \Leftarrow B$), new dependency type ($A \Leftrightarrow B$) - possible
Modify ($A \Leftarrow B$), new dependency type ($A \Rightarrow B$) - possible
Modify ($A \Leftarrow B$), new dependency type ($A \nLeftrightarrow B$) - possible
Modify ($A \Leftarrow B$), new dependency type ($A \bar{\wedge} B$) - possible (negated equivalence instead of NAND)
Modify ($A \Leftarrow B$), new dependency type ($A \vee B$) - possible
Modify ($A \Leftarrow B$), new dependency type ($A - B$) - possible

Modify ($A \prec_d B$), new dependency type ($A \prec B$) - possible (direct temporal dependency instead of eventual)

Modify ($A \prec_d B$), new dependency type ($A \succ B$) - possible (direct temporal dependency instead of eventual)

Modify ($A \prec_d B$), new dependency type ($A \succ_d B$) - possible

Modify ($A \prec_d B$), new dependency type ($A - B$) - possible

Combined change operations

Replace

Replace activity A - possible

Replace activity B - possible

Collapse

Collapse $\{A, B\}$ - possible

De collapse

De-collapse activity A (for any collapsed process) - possible

De-collapse activity B (for any collapsed process) - possible

Parallelize

Parallelize $\{A, B\}$ - possible

Swap

Swap activity A with B (or vice versa) - possible

Skip

Skip activity A - possible

Skip activity B - possible

Condition Update

Condition update, condition activity A , depending activity B - possible

Condition update, condition activity B , depending activity A - possible

Implementation of change operations for locked dependencies

Sequential process model We start with the most simple process model, a sequence of the activities A, B, C, D .

Locked dependency $A \prec_d B$

Delete activity A - possible

Delete activity C - possible

Insert activity X , condition: $C \prec X$ - possible

Collapse $\{A, B\}$ - possible

Collapse $\{B, C\}$ - possible

De-collapse A - possible

De-collapse C - possible

Replace A - possible

Replace C - possible

Skip activity A - possible

Skip activity C - possible

Swap activities A and B - Error: Temporal dependency from 'A' to 'B' is locked and cannot be changed.

Swap activities A and C - Error: Temporal dependency from 'A' to 'B' is locked and cannot be changed.

Swap activities C and D - possible Parallelize A and B - Error: Temporal dependency from 'A' to 'B' is locked and cannot be changed.

Parallelize B and C - Error: Temporal dependency from 'A' to 'B' is locked and cannot be changed.

Parallelize C and D - possible Condition update, A condition, B depending - possible

Condition update, A condition, C depending - possible

Condition update, C condition, D depending - possible

Locked dependency $A \Leftrightarrow B$

Delete activity A - Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.

Delete activity C - possible

Insert activity X , condition: $A \Leftrightarrow X$ - possible

Insert activity X , condition: $C \Leftrightarrow X$ - possible

Collapse $\{A, B\}$ - Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.

Collapse $\{B, C\}$ - Error: Activity 'B' cannot be deleted because existential dependency from 'A' to 'B' is locked.

De-collapse A - Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.

De-collapse C - possible

Replace A - Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.

Replace C - possible

Skip activity A - Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.

Skip activity C - possible

Swap activities A and B - possible

Swap activities A and C possible

Swap activities C and D - possible

Parallelize A and B - possible

Parallelize B and C - possible

Parallelize C and D - possible Condition update, A condition, B depending - possible

Condition update, A condition, C depending - possible

Condition update, C condition, D depending - possible

XOR split We continue with the process with the acceptance sequences $\{AB, AC\}$. Here we focus on cases, which we did not consider beforehand.

Locked dependency $A \Leftarrow B$

Parallelize A, B, C - Error: Existential dependency from 'A' to 'B' is locked and cannot be changed.